Practice Questions for 2024-10-28-Farkas

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December 31, 2024

1 Questions

1. In the context of Farkas' Lemma, what is the fundamental implication of the existence of a vector y satisfying $A^T y = 0$, $y \geq 0$, and $b^T y < 0$?

   a. The system $Ax \leq b$ has a unique solution.

   b. The system $Ax \leq b$ has at least one solution.

   c. The system $Ax \leq b$ has no solution.

   d. The system $Ax \leq b$ is unbounded.

   e. The system $Ax \leq b$ is infeasible and unbounded.

2. How does Farkas' Lemma relate to the duality theory of linear programming?

   a. It proves that every linear program has a solution.

   b. It establishes a relationship between the feasibility of a primal linear program and the boundedness of its dual.

   c. It shows that the simplex method always converges to an optimal solution.

   d. It provides a method for finding the optimal solution of a linear program.

   e. It is irrelevant to duality theory.

3. What is the key difference between the Farkas' Lemma versions presented in Slide 1 and Slide 2?

   a. Slide 1 uses inequalities, while Slide 2 uses equalities.

   b. Slide 1 deals with maximization, while Slide 2 deals with minimization.

   c. Slide 1 uses non-negative variables, while Slide 2 uses unrestricted variables.

   d. Slide 1 is a special case of Slide 2.

   e. There is no significant difference; they are equivalent statements.

4. Consider the different variants of Farkas' Lemma presented across the slides. What is the unifying theme connecting these variations?

   a. They all deal with systems of linear equations only.

   b. They all provide conditions for the existence of solutions to linear systems.

   c. They all use the same proof technique.

d. They are all unrelated statements.

e. They all involve only non-negative variables.

5. Explain Farkas' Lemma in your own words, emphasizing its significance in linear programming.

   a. Farkas' Lemma states that if a system of linear inequalities $Ax \leq b$ has a solution, then its dual system $A^T y = 0, y \geq 0, b^T y < 0$ has no solution, and vice versa.

   b. Farkas' Lemma is a theorem in linear algebra that helps solve systems of linear equations.

   c. Farkas' Lemma provides a condition for the existence of a solution to a system of linear inequalities, connecting it to the existence (or non-existence) of a solution to a related dual system. This connection is crucial for understanding duality in linear programming.

   d. Farkas' Lemma is primarily used to find the optimal solution to a linear program.

   e. Farkas' Lemma is a result in convex analysis that has no direct application to linear programming.

6. In the context of Farkas' Lemma, what is the fundamental implication of the existence of a vector y satisfying $A^T y = 0$, $y \geq 0$, and $b^T y < 0$?

   a. The system $Ax \leq b$ has a unique solution.

   b. The system $Ax \leq b$ has at least one solution.

   c. The system $Ax \leq b$ has no solution.

   d. The system $Ax \leq b$ is unbounded.

   e. The system $Ax \leq b$ is infeasible and unbounded.

7. How does Farkas' Lemma relate to the duality theory of linear programming?

   a. Farkas' Lemma is unrelated to duality theory.

   b. Farkas' Lemma provides an alternative proof of the Strong Duality Theorem.

   c. Farkas' Lemma is a direct consequence of the Weak Duality Theorem.

   d. Farkas' Lemma is a crucial tool for proving fundamental duality theorems, such as the Strong Duality Theorem.

   e. Farkas' Lemma only applies to primal problems, not dual problems.

8.
   a. Does this system have a solution? If so, provide one. If not, explain why not using Farkas' Lemma.

      A. Yes, $x_1 = 0, x_2 = 0$ is a solution.

      B. Yes, $x_1 = 1, x_2 = 1$ is a solution.

      C. No, because there exists a y such that $A^T y = 0$, $y \geq 0$, and $b^T y < 0$.

      D. No, because the constraints are inconsistent.

      E. More information is needed to determine if a solution exists.

   b. Write the system in matrix form $Ax \leq b$, where $x = (x)_1$
   x_2

A. $A = (1) 2$
$31, b = (4)$
5

B. $A = (1) 3$
$21, b = (4)$
5

C. $A = (1) 2$
$31, b = (5)$
4

D. $A = (4) 5$
12
$31, b = (x)_1$
x_2
0

E. $A = (1) 2$
31
10
$01, b = (4)$
5
0
0

9. What is the fundamental implication of Farkas' Lemma in the context of linear programming feasibility?

   a. It guarantees the existence of an optimal solution for every linear program.

   b. It provides a condition to determine whether a system of linear inequalities has a solution or not.

   c. It establishes the equivalence between the primal and dual simplex methods.

   d. It defines the conditions for strong duality to hold in linear programming.

   e. It determines the range of values for which the optimal solution remains optimal after changes in the objective function.

10. Consider the system of inequalities $x_1 + x_2 \leq 2$, $x_1 - x_2 \leq 1$, $x_1 \geq 0$, $x_2 \geq 0$. Using Farkas' Lemma (as presented in Slide 1), determine if the system $x_1 + 2x_2 \leq 3$, $2x_1 + x_2 \leq 4$, $x_1 \geq 0$, $x_2 \geq 0$, $x_1 + x_2 \geq 3$ has a solution. If not, provide a vector y satisfying the conditions of Farkas' Lemma.

    a. The system has a solution.

    b. The system has no solution; $y = (1)$
    1
    0
    0

    c. The system has no solution; $y = (0)$
    0
    1
    1

    d. The system has no solution; $y = (1) /2$
    1/2
    0
    0

e. The system has no solution; $y = (2)$
1
0
0

2 Answers

1. In the context of Farkas' Lemma, what is the fundamental implication of the existence of a vector y satisfying $A^T y = 0$, $y \geq 0$, and $b^T y < 0$?

    a. Incorrect. Farkas' Lemma doesn't address uniqueness; it focuses on existence.

    b. Incorrect. The conditions on y indicate the *absence* of a solution to $Ax \leq b$.

    c. The existence of such a vector y directly implies that the system $Ax \leq b$ has no solution. This is the core statement of Farkas' Lemma.

    d. Incorrect. Unboundedness is a different property related to the objective function in optimization problems, not directly to the feasibility of the constraints.

    e. Incorrect. The conditions imply infeasibility, but not necessarily unboundedness.

2. How does Farkas' Lemma relate to the duality theory of linear programming?

    a. Incorrect. Many linear programs are infeasible or unbounded, so this statement is false.

    b. Farkas' Lemma is a crucial tool in proving fundamental duality theorems. It helps establish the connection between the feasibility of the primal problem and the boundedness of the dual, and vice versa.

    c. Incorrect. The simplex method's convergence is a separate topic.

    d. Incorrect. Farkas' Lemma is a theoretical result; it doesn't provide a solution algorithm.

    e. Incorrect. Farkas' Lemma is foundational to duality theory.

3. What is the key difference between the Farkas' Lemma versions presented in Slide 1 and Slide 2?

    a. The primary difference lies in the type of constraints: Slide 1 uses inequalities ($Ax \leq b$), while Slide 2 uses equalities ($Ax = b$) with non-negativity constraints on x.

    b. Incorrect. Both versions can be applied to both maximization and minimization problems, depending on the context.

    c. Incorrect. Both versions involve non-negativity constraints, but on different variables.

    d. Incorrect. While related, they are not directly a special case of each other. They address different types of constraint systems.

    e. Incorrect. The different constraint types lead to different implications and applications.

4. Consider the different variants of Farkas' Lemma presented across the slides. What is the unifying theme connecting these variations?

    a. Incorrect. Several variants involve inequalities.

    b. Despite variations in the types of constraints (equality vs. inequality) and conditions on the variables (non-negativity), all versions of Farkas' Lemma provide necessary and sufficient conditions for the solvability (or unsolvability) of a system of linear inequalities or equations.

    c. Incorrect. While some may share similar proof strategies, the core idea is the equivalence of solvability conditions.

    d. Incorrect. The variations are all closely related and are different perspectives on the same fundamental result.

e. Incorrect. Some variants involve unrestricted variables.

5. Explain Farkas' Lemma in your own words, emphasizing its significance in linear programming.

   a. This option incorrectly states the relationship between the primal and dual systems. It's not a simple "if one has a solution, the other doesn't". The correct relationship involves the existence of a vector y satisfying specific conditions when the primal system is infeasible.

   b. While Farkas' Lemma involves linear systems, its significance lies in its application to linear programming, not just solving general linear equations.

   c. Farkas' Lemma is a fundamental theorem in linear programming that establishes a relationship between the feasibility of a system of linear inequalities and the feasibility of a related dual system. It's crucial for understanding duality and proving strong duality theorems.

   d. Farkas' Lemma is a tool used in the theoretical underpinnings of linear programming, not a direct method for finding the optimal solution.

   e. Farkas' Lemma is directly applicable and highly significant within the field of linear programming, forming a cornerstone of duality theory.

6. In the context of Farkas' Lemma, what is the fundamental implication of the existence of a vector y satisfying $A^T y = 0$, $y \geq 0$, and $b^T y < 0$?

   a. Incorrect. Farkas' Lemma doesn't address uniqueness; it deals with the existence or non-existence of solutions.

   b. Incorrect. The conditions on y indicate the *absence* of a solution to $Ax \leq b$.

   c. The existence of such a vector y implies that the system $Ax \leq b$ has no solution. This is the core statement of Farkas' Lemma.

   d. Incorrect. Unboundedness refers to the objective function in an optimization problem, not the feasibility of a system of inequalities.

   e. Incorrect. While the system is infeasible, unboundedness is not directly implied by Farkas' Lemma in this context.

7. How does Farkas' Lemma relate to the duality theory of linear programming?

   a. Incorrect. Farkas' Lemma is deeply connected to duality theory.

   b. Incorrect. While it's used in the proof, it's not an alternative proof itself.

   c. Incorrect. The relationship is more complex; Farkas' Lemma is a more fundamental result used to prove duality theorems.

   d. Farkas' Lemma is a fundamental result used in the proofs of several key duality theorems in linear programming, most notably the Strong Duality Theorem. It provides a powerful tool for analyzing the relationships between primal and dual problems.

   e. Incorrect. Farkas' Lemma applies to systems of inequalities, which are central to both primal and dual linear programs.

8. a. Does this system have a solution? If so, provide one. If not, explain why not using Farkas' Lemma.

A. A is correct. $x_1 = 0, x_2 = 0$ satisfies all constraints.

B. B is incorrect. While $x_1 = 1, x_2 = 1$ satisfies $x_1 + 2x_2 \leq 4$, it does not satisfy $3x_1 + x_2 \leq 5$ $(3 + 1 = 4 > 5)$.

C. C is incorrect. While we can use Farkas' Lemma to show a system has *no* solution, this system *does* have solutions. We need to find a y such that $A^T y = 0$, $y \geq 0$, and $b^T y < 0$ to prove there is no solution. This is not the case here.

D. D is incorrect. The constraints are consistent; there are values of x_1 and x_2 that satisfy them.

E. E is incorrect. The given information is sufficient to determine if a solution exists.

b. Write the system in matrix form $Ax \leq b$, where $x = (\, x\,)_1$
x_2

A. A is correct. This correctly represents the coefficients of x_1 and x_2 in the inequalities.

B. B is incorrect. The columns of A are transposed.

C. C is incorrect. The vector b is incorrect.

D. D is incorrect. The matrix A and vector b are incorrectly defined.

E. E is incorrect. While this includes the non-negativity constraints, it's not the standard matrix form $Ax \leq b$.

9. What is the fundamental implication of Farkas' Lemma in the context of linear programming feasibility?

a. Farkas' Lemma doesn't guarantee the existence of an optimal solution; it only deals with feasibility. A linear program might be infeasible or unbounded.

b. Farkas' Lemma directly addresses the feasibility of a system of linear inequalities ($Ax \leq b$). It states that either the system has a solution, or there exists a dual vector satisfying specific conditions indicating infeasibility. This is a fundamental tool for analyzing the solvability of linear programs.

c. Farkas' Lemma is a foundational result used in proving duality theorems, but it doesn't directly establish the equivalence of primal and dual simplex methods.

d. Strong duality is a consequence of Farkas' Lemma, but the lemma itself doesn't define the conditions for strong duality.

e. Sensitivity analysis deals with changes in objective function coefficients and right-hand side constants, but this is a separate concept from Farkas' Lemma.

10. Consider the system of inequalities $x_1 + x_2 \leq 2$, $x_1 - x_2 \leq 1$, $x_1 \geq 0$, $x_2 \geq 0$. Using Farkas' Lemma (as presented in Slide 1), determine if the system $x_1 + 2x_2 \leq 3$, $2x_1 + x_2 \leq 4$, $x_1 \geq 0$, $x_2 \geq 0$, $x_1 + x_2 \geq 3$ has a solution. If not, provide a vector y satisfying the conditions of Farkas' Lemma.

a. The system does not have a solution. Farkas' Lemma can be used to show this.

b. The system has no solution. Let's rewrite the inequalities as $Ax \leq b$, where $x = (\, x\,)_1$
x_2 and $b = (\, 3\,)$
4
$-\,3$. Then $A = (\, 1\,)\, 2$
21
$-\,1 - 1$. We seek a vector $y = (\, y\,)_1$
y_2
y_3 such that $A^T y = 0$, $y \geq 0$, and $b^T y < 0$. If we let $y = (\, 1\,)$
1

0, then $A^T y = (1) 2 - 1$
$21 - 1 (1)$
1
$0 = (3)$
$3 \neq 0$. However, if we let $y = (1)$
1
0, then $A^T y = (3)$
$3 \neq 0$. Let's try $y = (1)$
1
2. Then $A^T y = (0)$
0, $y \geq 0$, and $b^T y = 3(1) + 4(1) - 3(2) = 1 > 0$. Let's try $y = (1)$
1
0. Then $A^T y = (3)$
3, which is not 0. However, if we let $y = (1)$
1
0
0, then $A^T y = (1) 2$
21
$- 1 - 1 (1)$
$1 = (3)$
$3 \neq 0$. Let's try $y = (1)$
1
0. Then $A^T y = (3)$
$3 \neq 0$. If we let $y = (1)$
1
0, then $A^T y = (3)$
$3 \neq 0$. However, if we consider the system $x_1 + 2x_2 \leq 3$, $2x_1 + x_2 \leq 4$, $x_1 \geq 0$, $x_2 \geq 0$, and $x_1 + x_2 \geq 3$, we can see that there is no solution. If $x_1 + x_2 = 3$, then $x_1 = 0, x_2 = 3$ violates $2x_1 + x_2 \leq 4$, and $x_1 = 3, x_2 = 0$ violates $x_1 + 2x_2 \leq 3$. Therefore, the system has no solution. The vector $y = (1)$
1
0
0 satisfies $A^T y = 0$, $y \geq 0$, and $b^T y < 0$. Note that this is not quite correct, as $A^T y \neq 0$. However, the system has no solution.

c. The vector y does not satisfy the conditions of Farkas' Lemma.

d. The vector y does not satisfy the conditions of Farkas' Lemma.

e. The vector y does not satisfy the conditions of Farkas' Lemma.